Week 8 Project — Time Series Forecasting

Project Overview

This project forecasts daily retail sales using historical sales data and machine learning models.

Prediction Task

- **Track:** A — Time Series Forecasting
- **Target:** `sales`
- **Forecast horizon:** One day ahead
- **Forecast refresh:** Daily
- **Baseline:** Seasonal Naive, using sales from seven days earlier
- **Evaluation metrics:** MAE and RMSE

Dataset

The project uses `retail_sales.csv`.

The dataset contains daily sales observations. The exact date range, number of records, missing values, and other columns are documented in the notebook.

The dataset may not contain all external factors that influence sales, such as promotions, holidays, and changing customer behavior.

Models

1. Seasonal Naive baseline
2. Random Forest Regressor
3. Histogram-based Gradient Boosting Regressor

The machine learning models are compared using expanding-window time-series validation.

Evaluation Protocol

The latest 56 daily observations are reserved for final testing. Earlier observations are used for training and validation.

The final test set is not used for hyperparameter selection. Features are constructed using historical sales and calendar information available at forecast time.

Metrics

- **MAE:** Average absolute difference between actual and predicted sales.
- **RMSE:** Square root of the average squared prediction error.

Lower values indicate better forecasting accuracy.

Key Findings

Add the actual baseline and final model MAE and RMSE from the notebook. State whether the selected model outperformed the baseline and identify any weak weekdays or periods.

Production Plan

Forecasts are generated daily using batch processing. Data quality, feature freshness, forecast errors, and changes in demand are monitored. The seasonal-naive model serves as a fallback if the main forecasting model fails.

How to Run

1. Install the dependencies:

```
pip install -r requirements.txt
```

2. Place `retail_sales.csv` in the project folder.
3. Open Jupyter Notebook.
4. Run `Week8_TimeSeries_Forecasting.ipynb` from the first cell to the last.

Reproducibility and Limitations

The notebook documents the chronological split, validation procedure, feature construction, model selection, and final evaluation.

Results represent one held-out test period and may not generalize to all future demand conditions.

Dataset Access and License

Document the original dataset source, applicable license, and any manual download steps here.